

First-pass battery estimate

I am doing a **first-pass battery estimate**, which means this is an **engineering estimate** to help choose a battery now, before having final measurements. It will help decide two things: the **minimum battery capacity** needed for 6 hours, and the **minimum discharge current capability** the battery should have so it can safely power the system.

What we know so far

We know that the design is using a **one-cell lithium iron phosphate battery**, and that we want to generate **5 volts, 3.3 volts, and 1.8 volts**. We also know we are using an **ESP32 development board**, an **STM32F103**, and **TMP75B temperature sensors**, and that communication will be **Bluetooth Low Energy** with periodic updates every 5 to 15 minutes.

What we do not know yet

We do not yet know the exact current on each rail in the real system, especially the **5 volt rail current** for the ESP32 development board. We also do not know exactly what is on the **1.8 volt rail** and how much current it draws. Because of that, we have to make reasonable assumptions for now.

Step 1: Make a first-pass current budget (by rail)

To estimate the battery, we first estimate how much current your system draws on each rail, both on average and at peak moments.

For this first-pass estimate, I assumed the **5 volt rail** (which powers the ESP32 development board through its board power path) draws about **120 milliamps on average** and can have **500 milliamp peaks**. I chose this because development boards usually draw more current than bare modules, and they can have short bursts when the radio is active.

For the **3.3 volt rail**, I assumed **20 milliamps average** and **60 milliamps peak**. This is a placeholder for the STM32 and any small logic on that rail.

For the **1.8 volt rail**, I assumed **10 milliamps average** and **30 milliamps peak** because we do not yet know the exact 1.8 volt load.

The **TMP75B temperature sensors** draw very little current compared to the microcontrollers, so they are effectively already covered inside these small rail estimates and do not dominate the battery choice.

Step 2: Estimate average system power (what the system uses over time)

Now we convert those average rail currents into **average power**, because batteries are most cleanly sized using power and energy.

For the **5 volt rail**, the average power is:

- $5 \text{ volts} \times 0.120 \text{ amps} = \mathbf{0.600 \text{ watts}}$

For the **3.3 volt rail**, the average power is:

- $3.3 \text{ volts} \times 0.020 \text{ amps} = \mathbf{0.066 \text{ watts}}$

For the **1.8 volt rail**, the average power is:

- $1.8 \text{ volts} \times 0.010 \text{ amps} = \mathbf{0.018 \text{ watts}}$

When we add those together, the **total average output power** is:

- $0.600 + 0.066 + 0.018 = \mathbf{0.684 \text{ watts}}$

So, the first-pass estimate for your average system power is **0.684 watts**.

Step 3: Convert average system power into average battery current

Even though the rails are 5 volts, 3.3 volts, and 1.8 volts, the battery is a **one-cell lithium iron phosphate battery**, so the battery current is not the same as the rail currents. The DC/DC converters change voltage, and they are not perfectly efficient.

For a first-pass estimate, I assumed an **average battery voltage of 3.2 volts** and an **overall converter efficiency of 80 percent**.

The formula is:

- $\text{average battery current} = \text{total average output power} \div (\text{efficiency} \times \text{average battery voltage})$

So we compute:

- $\text{efficiency} \times \text{average battery voltage} = 0.80 \times 3.2 = \mathbf{2.56}$
- $\text{average battery current} = 0.684 \div 2.56 = \mathbf{0.267 \text{ amps}}$ (approximately)

That means the estimated **average battery current** is about **267 milliamps**.

Step 4: Compute the minimum battery capacity for 6 hours

Now we use the average battery current to calculate how much capacity you need to run for 6 hours.

The raw capacity calculation is:

- capacity = average battery current × time
- capacity = 0.267 amps × 6 hours = **1.603 amp-hours**

So the raw minimum (with no margin at all) would be about **1.603 amp-hours**, which is **1603 milliamp-hours**.

Adding realistic margin (very important)

In real designs, you should not size the battery using the raw number only, because:

- you usually do not want to plan on using 100 percent of the battery capacity,
- your estimates can be off,
- converter efficiency changes with load,
- battery performance changes with age and temperature.

So I added two margins:

- First, I assumed you only want to use **80 percent** of the battery capacity (to avoid planning for full depletion every cycle).
- Then I added an additional **20 percent design margin** for uncertainty and real-world losses.

When those margins are applied, the recommended first-pass battery capacity becomes about **2.405 amp-hours**, which is about **2405 milliamp-hours**.

So the **first-pass battery capacity target** is roughly **2400 milliamp-hours minimum**, and choosing something larger gives you more safety and runtime margin.

Step 5: Estimate peak system power (for discharge current capability)

Battery capacity determines runtime, but the battery also has to be able to supply **peak current** safely. To estimate that, we use the peak currents on each rail.

For the **5 volt rail**, the peak power is:

- 5 volts $\times$ 0.500 amps = **2.500 watts**

For the **3.3 volt rail**, the peak power is:

- 3.3 volts $\times$ 0.060 amps = **0.198 watts**

For the **1.8 volt rail**, the peak power is:

- 1.8 volts $\times$ 0.030 amps = **0.054 watts**

Adding those gives a **total peak output power** of:

- $2.500 + 0.198 + 0.054 =$ **2.752 watts**

So the first-pass estimate for peak output power is **2.752 watts**.

Step 6: Convert peak output power into peak battery current

To estimate peak battery current, we use a more stressful battery condition because peaks often matter most when the battery is lower.

For this estimate, I assumed:

- a **low battery voltage of 2.8 volts** (conservative),
- and **85 percent converter efficiency** during peak loading.

The formula is:

- peak battery current = peak output power $\div$ (efficiency $\times$ low battery voltage)

So we compute:

- efficiency $\times$ low battery voltage = $0.85 \times 2.8 =$ **2.38**
- peak battery current = $2.752 \div 2.38 =$ **1.156 amps** (approximately)

That means the estimated **peak battery current** is about **1.2 amps**.

Step 7: What battery specs this points to (first-pass targets)

Based on this first-pass estimate, the battery should be chosen to meet both runtime and current delivery needs.

For battery capacity

The estimate suggests a minimum of about **2400 milliamp-hours** after adding margin. Because some of the rail loads are still uncertain (especially the 5 volt rail and 1.8 volt rail), a more comfortable choice would be something in the **3000 to 5000 milliamp-hour** range.

For battery discharge capability

Even though the estimated peak battery current is only about **1.2 amps**, we should still choose a battery with margin.

A good first-pass target is:

- **at least 2 amps maximum continuous discharge current**, and
- if pulse current is listed, **at least 3 amps pulse current** is a nice comfortable minimum.

If a battery datasheet does not list pulse current, that's ok because I can focus on the **maximum continuous discharge current** and choose a battery with extra margin.

Step 8: What will most likely change this estimate later

This estimate is useful, but it will improve a lot once the actual current is measured. The biggest things that can change the result are:

- the **actual average current on the 5 volt rail** (this is probably the biggest factor),
 - what exactly is on the **1.8 volt rail** and how often it is active,
 - how efficiently the ESP32 firmware uses Bluetooth Low Energy power-saving features,
 - and the actual efficiencies of the converters chosen.
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Final takeaway

Right now, this first-pass estimate is good enough to help pick a battery candidate. The system probably needs **more than 1000 milliamp-hours**, and a **one-cell lithium iron phosphate battery in the 3000 to 5000 milliamp-hour range** is a much safer starting point for a 6-hour wearable prototype. Also the battery should be able to supply at least **about 2 amps continuously** so it can comfortably handle peak demand with margin.

Chosen Battery: **LiFePO4 26700 Rechargeable Cell: 3.2V 5000 mAh (16Wh, 10A/2C Rate)**
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